

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : AI

Aim: Association Rule Mining

Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.

Generate association rules from the frequent itemsets and calculate their support and confidence.

Interpret and analyze the discovered association rules.

Input

```
PRACTICAL 9.py - C:\Users\DELL\Downloads\PRACTICAL 9.py (3.13.11)
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import pandas as pd
import matplotlib.pyplot as plt

from mlxtend.frequent_patterns import apriori
from mlxtend.frequent_patterns import association_rules

# Load dataset
df = pd.read_csv("saleshourly.csv")

print("Dataset shape:", df.shape)

print("nColumns:")
print(df.columns)

# Drug columns
drug_columns = [
    "M01AB",
    "M01AE",
    "N02BA",
    "N02BE",
    "N05B",
    "N05C",
    "R03",
    "R06"
]

# Keep only drug columns
transactions = df[drug_columns].copy()

# Convert sales quantities into 0 and 1
# 1 means the drug was sold during that hour
# 0 means the drug was not sold
transactions = transactions > 0

# Remove hours where no drug was sold
transactions = transactions[transactions.sum(axis=1) > 0]
```

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```
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print("\nNumber of transactions:", len(transactions))

# Find frequent itemsets
frequent_itemsets = apriori(
    transactions,
    min_support=0.05,
    use_colnames=True
)

# Sort by support
frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("\nFrequent Itemsets:")
print(frequent_itemsets)

# Generate association rules
rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.30
)

# Keep useful columns
rules = rules[
    [
        "antecedents",
        "consequents",
        "support",
        "confidence",
        "lift"
    ]
]

# Sort rules by confidence
rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("\nAssociation Rules:")
print(rules)

# Display rules in an easier format
print("\nTop Association Rules:")

for index, row in rules.head(10).iterrows():
    antecedent = " ".join(row["antecedents"])
    consequent = " ".join(row["consequents"])

    print(
        antecedent,
        "->",
        consequent,
        "\nSupport:",
        round(row["support"], 3),
        "\nConfidence:",
        round(row["confidence"], 3),
        "\nLift:",
        round(row["lift"], 3)
    )

# Graph of top 10 rules by confidence
top_rules = rules.head(10).copy()

rule_names = []

for index, row in top_rules.iterrows():
    antecedent = " ".join(row["antecedents"])
```

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```
# Graph of top 10 rules by confidence
top_rules = rules.head(10).copy()
```

```
rule_names = []
```

```
for index, row in top_rules.iterrows():
```

```
    antecedent = " ".join(row["antecedents"])
    consequent = " ".join(row["consequents"])
```

```
    rule_names.append(
        antecedent + " -> " + consequent
    )
```

```
plt.figure(figsize=(9, 6))
```

```
plt.barh(
    rule_names[:-1],
    top_rules["confidence"].values[:-1]
)
```

```
plt.title("Top Association Rules by Confidence")
plt.xlabel("Confidence")
plt.ylabel("Association Rule")
```

```
plt.tight_layout()
plt.show()
```

Output

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Dataset shape: (50532, 13)

Columns:

Index(['datum', 'M01AB', 'M01AE', 'N02BA', 'N02BE', 'N05B', 'N05C', 'R03',
 'R06', 'Year', 'Month', 'Hour', 'Weekday Name'],
 dtype='object')

Number of transactions: 26588

Frequent Itemsets:

support	itemsets
3 0.767339	(N02BE)
4 0.397096	(N05B)
1 0.357868	(M01AE)
0 0.334700	(M01AB)
19 0.294080	(N02BE, N05B)
2 0.280051	(N02BA)
13 0.276064	(N02BE, M01AE)
9 0.253498	(N02BE, M01AB)
16 0.214345	(N02BA, N02BE)
6 0.198774	(R06)
21 0.147360	(N02BE, R06)
14 0.141981	(N05B, M01AE)
10 0.137919	(M01AB, N05B)
5 0.128329	(R03)
7 0.127764	(M01AB, M01AE)
17 0.119866	(N02BA, N05B)
30 0.112231	(N05B, N02BE, M01AE)
27 0.106401	(N02BE, M01AB, N05B)
12 0.105235	(N02BA, M01AE)
24 0.103317	(N02BE, M01AB, M01AE)
8 0.101474	(N02BA, M01AB)
20 0.098390	(R03, N02BE)
32 0.095156	(N02BA, N02BE, N05B)
29 0.085866	(N02BA, N02BE, M01AE)
26 0.081766	(N02BA, N02BE, M01AB)
23 0.075937	(R06, N05B)

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```
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15 0.075034 (R06, M01AE)
11 0.069505 (R06, M01AB)
31 0.059162 (N02BE, M01AE, R06)
33 0.057545 (N02BE, N05B, R06)
25 0.057432 (N05B, M01AB, M01AE)
18 0.056793 (N02BA, R06)
28 0.054122 (N02BE, M01AB, R06)
22 0.053821 (R03, N05B)

Association Rules:
antecedents consequents support confidence lift
37 (N02BA, M01AE) (N02BE) 0.085866 0.815940 1.063337
27 (M01AB, M01AE) (N02BE) 0.103317 0.808655 1.053843
41 (N02BA, M01AB) (N02BE) 0.081766 0.805782 1.050100
33 (N02BA, N05B) (N02BE) 0.095156 0.793850 1.034550
17 (N05B, M01AE) (N02BE) 0.112231 0.790464 1.030137
47 (R06, M01AE) (N02BE) 0.059162 0.788471 1.027540
54 (R06, M01AB) (N02BE) 0.054122 0.778680 1.014780
22 (M01AB, N05B) (N02BE) 0.106401 0.771475 1.005391
3 (M01AE) (N02BE) 0.276064 0.771414 1.005310
31 (R03) (N02BE) 0.098390 0.766706 0.999175
6 (N02BA) (N02BE) 0.214345 0.765377 0.997444
49 (R06, N05B) (N02BE) 0.057545 0.757801 0.987570
5 (M01AB) (N02BE) 0.253498 0.757388 0.987033
7 (R06) (N02BE) 0.147360 0.741343 0.966123
1 (N05B) (N02BE) 0.294080 0.740576 0.965123
50 (M01AB, M01AE) (N05B) 0.057432 0.449514 1.132003
32 (N02BA, N02BE) (N05B) 0.095156 0.443938 1.117959
14 (N02BA) (N05B) 0.119866 0.428015 1.077862
20 (N02BE, M01AB) (N05B) 0.106401 0.419733 1.057005
55 (R03) (N05B) 0.053821 0.419402 1.056172
52 (M01AB, N05B) (M01AE) 0.057432 0.416417 1.163603
10 (M01AB) (N05B) 0.137919 0.412069 1.037705
25 (N02BE, M01AB) (M01AE) 0.103317 0.407567 1.138874
16 (N02BE, M01AE) (N05B) 0.112231 0.406540 1.023780
51 (N05B, M01AE) (M01AB) 0.057432 0.404503 1.208555
46 (N02BE, R06) (M01AE) 0.059162 0.401480 1.121866

Ln: 52 Col: 7

Top Association Rules:
N02BA, M01AE -> N02BE | Support: 0.086 | Confidence: 0.816 | Lift: 1.063
M01AB, M01AE -> N02BE | Support: 0.103 | Confidence: 0.809 | Lift: 1.054
N02BA, M01AB -> N02BE | Support: 0.082 | Confidence: 0.806 | Lift: 1.05
N02BA, N05B -> N02BE | Support: 0.095 | Confidence: 0.794 | Lift: 1.035
N05B, M01AE -> N02BE | Support: 0.112 | Confidence: 0.79 | Lift: 1.03
R06, M01AE -> N02BE | Support: 0.059 | Confidence: 0.788 | Lift: 1.028
R06, M01AB -> N02BE | Support: 0.054 | Confidence: 0.779 | Lift: 1.015
M01AB, N05B -> N02BE | Support: 0.106 | Confidence: 0.771 | Lift: 1.005
M01AE -> N02BE | Support: 0.276 | Confidence: 0.771 | Lift: 1.005
R03 -> N02BE | Support: 0.098 | Confidence: 0.767 | Lift: 0.999

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```

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Figure 1

